

Programming Exam 6

Date/Time: 2026.05.05 08:00 – 08:50

(程式檔命名學號_pe6.py，上傳至 Moodle PE6 上傳區)

Problem: Shortest Healthy Walking Streak

A health app records how many steps a person walks each day. The doctor says that a person should accumulate at least T steps within **one consecutive block of days** to count as an effective recovery streak. Your job is to find that block. Given the number of steps walked over N days, find the shortest consecutive period whose total number of steps is at least T . If there are multiple answers, use the following rules:

- (1) Choose the one with the **fewest days**.
- (2) If there is still a tie, choose the one with the **smaller total number of steps**.
- (3) If there is still a tie, choose the one that **starts earlier**.

If no consecutive period can reach T steps, print **IMPOSSIBLE**.

Input Format

- The 1st line contains two integers N and T
- The 2nd line has N integers, where the i -th integer is the number of steps walked on day i
- Constraints: $1 \leq N \leq 200$, $0 \leq \text{steps}[i] \leq 50000$, $1 \leq T \leq 10^9$

Output Format

If a valid period exists, print three integers: starting day, ending day, and total number of steps

Use **1-based indexing** for the day numbers. If no valid period exists, print: **IMPOSSIBLE**

Example Input

8 15000

3200 4500 2800 7000 3500 4000 2000 6000

Example Output

5 8 15500

Explanation

- Days 5 to 8 have total steps = $3500 + 4000 + 2000 + 6000 = 15500$
- No shorter consecutive period can reach 15000 steps
- So this is the correct answer

Sample Input/Output

(以下是你程式執行後須印出的結果)

```
c:\workspace>python pe6.py
8 15000
3200 4500 2800 7000 3500 4000 2000 6000
5 8 15500
```

```
c:\workspace>python pe6.py
6 5000
1200 5300 2000 1000 4000 3000
2 2 5300
```

```
c:\workspace>python pe6.py
5 20
3 4 2 5 6
1 5 20
```

```
c:\workspace>python pe6.py
5 30
3 4 2 5 6
IMPOSSIBLE
```

```
c:\workspace>python pe6.py
4 10
6 5 4 6
3 4 10
```

```
c:\workspace>python pe6.py
5 9
4 5 4 5 1
1 2 9
```

```
c:\workspace>python pe6.py
7 7
0 0 3 0 4 0 0
3 5 7
```

```
c:\workspace>python pe6.py
1 7
7
1 1 7
```

```
c:\workspace>python pe6.py
1 8
7
IMPOSSIBLE
```

```
c:\workspace>python pe6.py
7 11
2 2 2 2 5 6 1
5 6 11
```

```
c:\workspace>python pe6.py
5 8
9 8 10 8 12
2 2 8
```

```
c:\workspace>python pe6.py
5 8
8 3 8 10 8
1 1 8
```

```
c:\workspace>python pe6.py
6 9
2 2 2 3 3 6
5 6 9
```

```
c:\workspace>python pe6.py
6 6
1 2 6 3 4 5
3 3 6
```

(繳交是交 pe6.py 檔，不是交截圖)

Note: You need to write comments (註解) for each part in your code.